TEAM 1: THE HASANOV SYNDICATE

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*ES51 - Fall 2024*

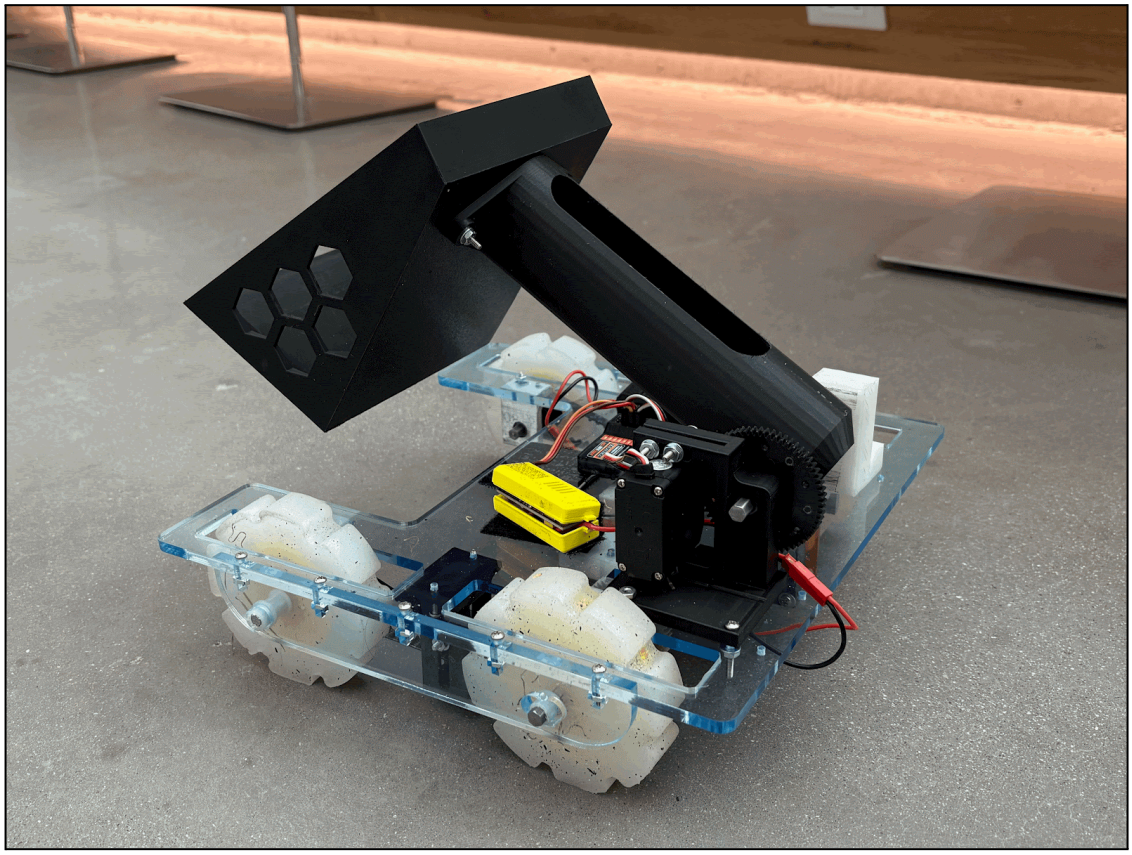


**Figure 1** - Bumblebee

# Abstract

The goal of this project was to design a robot to compete in the Turf Wars competition. The most important requirement was fitting within a 11” x 11” x 11” box (Box of Justice), and the challenges available for scoring were: collecting and storing 6 ping pong balls, going up, down, and under a 30° ramp, collecting dog toys and hockey pucks and deposit those in boxes located in the perimeter of the arena, ringing a bell, and a stacking challenge involving a wooden ring, ball, and cube. Due to time and material constraints, our group went for an all-purpose type of solution that was able to tackle every task, except for the stacking challenge. We focused on going up and down the ramp and collecting and storing the ping pong balls. In order to complete these tasks, our design was an all-purpose, 3D printed scoop with a built-in storage in the arm, both controlled by a single servo and a hex shaft held by a delrin support. To help us make navigation more intuitive, we attached rubber bands to the remote control and inverted the polarity of a couple of components.We optimized our robot for endurance, and though the team didn’t win the competition we managed to complete two rounds with no technical failures.

# Concept Development

### Constraints

- Size constraints: 11”x11”x11”;
- Time constraints: 4 minutes for challenges during competition, roughly 10 weeks to design, fabricate, and test the robot;
- Material constraints: Robot must be fabricated using only the materials available in the ES51 lab. A maximum of 4 sheets of acrylic could be used;
- Technical constraints: The robot must be able to perform tasks using a maximum of 6 actuators, including servos and DC motors. The robot must also have components fabricated using all different techniques learned in class, such as milling and lathing.

### First sketches and Design Review 1

Initially, the challenges in the competition rules provided a loophole in which no limit was placed on the number of points our team could gain by climbing the center 30° ramp. As such our strategy seemed clear: design a robot that could climb the ramp as quickly and reliably as possible. Thus, our preliminary designs focused on optimizing the lowest center of mass possible, high traction, and an inability to be “killed” (losing a “life” because of tilting) through the use of treads, different chassis shapes, and even spikes as pictured below.

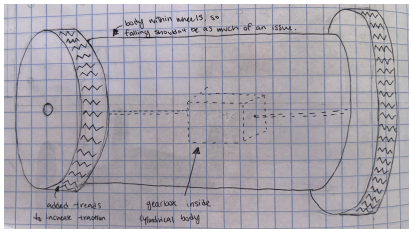


**Figure 2.1** - Roller

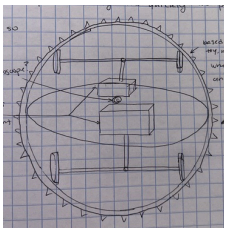


**Figure 2.2 -** Sphero

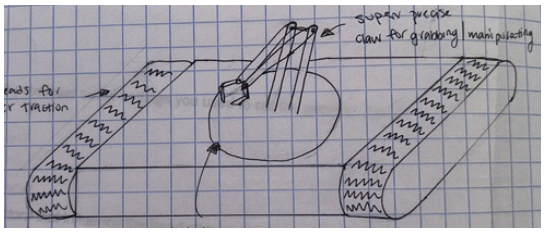


**Figure 2.3** - Grabber

However, our plans were swiftly ruined as the rules changed, limiting the number of climbs to two, disrupting the core of our game strategy and forcing us to restart our ideation process. We again began by developing a new strategy, ultimately focusing on a more well-rounded design that could accomplish climbing and going under the ramp efficiently and consistently, picking up and storing ping pong balls, and depositing pucks and dog toys. While we wanted our robot to be able to ring the bell, we didn't place that as a top priority since we figured we could simply roll over the bell to score. Our team decided to forgo trying to accomplish the stacking challenge since we thought the number of points was not worth the added dexterity we'd have to implement.

With our strategy clarified, we then began developing different concepts that could accomplish each challenge. Our highest priority was accomplishing the ramp tasks given how many points we could gain relatively easily. As such, we approached our new designs with many similar characteristics as our original designs, aiming to design robots with a low center of mass, high traction, and good survivability. Our first concept focused almost exclusively on collecting ping pong balls, simplifying our strategy with a door-like enclosure system. However, although this would excel at ball collection and hill challenges as well as being low-cost and mobile, it also sacrifices the higher point tasks we originally hoped to achieve and lacks a secure ball carriage. Our second design featured a very stable lift system that could be used to scoop and raise up dog toys and pucks, as well as a robust ping pong ball container. But, given the tall height and thus higher center of mass, we had concerns about the hill climbing ability of the robot, as well as the potential difficulties of depositing collected items. Our final design implemented a similar scoop-lift mechanism with an added tilt function to both better deposit items as well as tilt ping pong balls back into the more secure carriage system. To maximize our scooping area this design came with the sacrifice of two front wheels, leading to concerns over mobility and careful tolerancing.

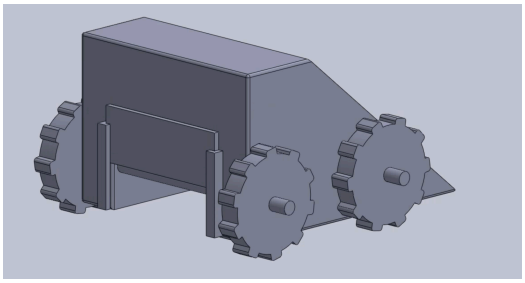


**Figure 3.1** - Ball Grabber

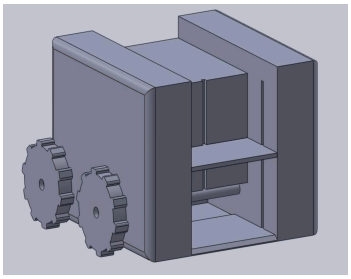


**Figure 3.2** - Mobile D.A.

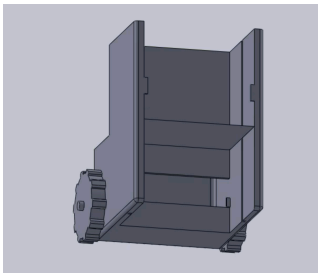


**Figure 3.3** - Tilt Dual

We used a Pugh Matrix to compare the merits of each concept based on the criteria of height, ability to climb hills, storing ping pong balls, picking up and dropping off objects, and finally cost, weighing each of them based on how important we found each characteristic. Adding up the scores, the Dual Action design came out as the best concept that we then protyped out of cardboard to better understand and confirmed the functionality that we hoped to achieve from it.

| Criteria | Baseline | Weight | Ball Grabber | Dual Action | Tilt DA |
|---|---|---|---|---|---|
| Height | 0 | 1 | +1 (+1) | -1 (-1) | -1 (-1) |
| Incline Mobility | 0 | 4 | +2 (+8) | -1 (-4) | 0 (0) |
| Ping Pong Storage | 0 | 3 | +1 (+3) | -1 (-3) | 0 (0) |
| Pick Up Mechanism | 0 | 4 | -2 (-8) | +1 (+4) | +3 (+12) |
| Drop Off Mechanism | 0 | 4 | 0 (0) | +1 (+4) | +2 (+8) |
| Cost | 0 | 2 | +2 (+4) | -1 (-2) | +1 (+2) |
| Total | - | - | 8 | -2 | 21 |

**Figure 4** - DR1 Pugh Matrix



**Figure 5** - Dual Action Prototype

## Design Review 2

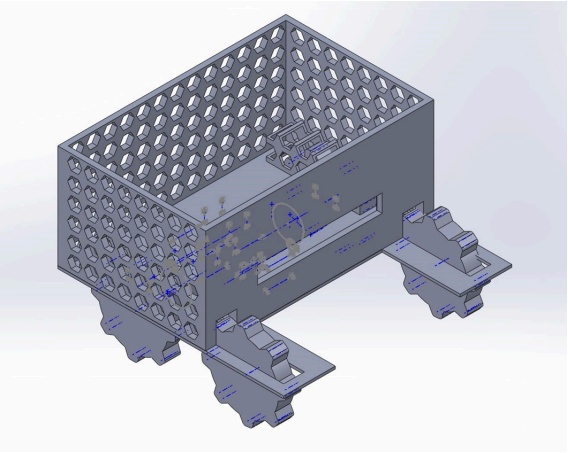


**Figure 6** - DR2 CAD

We went into Design Review 2 with the above CAD model for our robot, with a hexagonal pattern to remove material and undesired weight from the robot (which is where the name “The Bumblebee” was derived from), as well as a protective skirt around the wheels to prevent obstacles from obstructing the robot’s movement as recommended in Design Review 1. We also updated our Pugh Matrix as follows:

| Criteria | Baseline | Weight | Tilt D.A |
|---|---|---|---|
| Height | 0 | 3 | -1 (-2) |
| Weight | 0 | 3 | +2 (6) |
| Storage | 0 | 3 | 0 (0) |
| Pick up mechanism | 0 | 4 | +3 (+12) |
| Drop-off mechanism | 0 | 4 | +2 (+8) |
| Control simplicity | 0 | 2 | +1 (+2) |
| Total | - | - | **26** |

**Figure 7** - DR2 Pugh Matrix

Although we had a strong design for the body of our robot, we were struggling to fully ideate our lifting mechanism, which involved a platform that could be lifted in the vertical axis while also being able to angle itself with a servo. Moreover, following this second design review we got great feedback on our overall design but were told that our lifting mechanism was simply unrealistic to design due to the need to integrate both a servo motor and a mechanism to move the platform upwards and downwards. So, we went back to the drawing board and decided on a whole new lifting mechanism based on advice from Professor Hasanov. We came up with an all-in-one mechanism that scoops all of the items that we need to interact with, stores ping pong balls in an internal deposit within the arm, and moves dog toys and hockey pucks into their respective boxes by rotating the arm 180 degrees, as powered by a servo with a torque focused gear ratio. This design became the basis for our final robot design.

## Analysis

**Robot tippage, wheel torque, and arm torque calculations**

### 1 Calculating Forces in the Y-direction on a Flat Plane

$$F_{net_y} = ma$$
$$F_{net_y} = 0$$
$$mg - F_N = 0$$
$$mg = F_N$$
$$2.45kg \times 9.81m/s = F_N$$
$$24.03N = F_N$$

### 2 Calculating Forces in the Y-direction on a $30^\circ$ Plane

$$F_{net_y} = ma$$
$$F_{net_y} = 0$$
$$mg_y - F_N = 0$$
$$mg \times sin(30^\circ) = F_N$$
$$2.45kg \times 9.81m/s \times sin(30^\circ) = F_N$$
$$12.02N = F_N$$

## 3 Calculating Necessary Wheel Torque. Assuming Top-speed of 0.15m/s Achieved Within 1s

$$mg_x = mgcos(30^\circ) = 2.45kg \times 9.81m/s \times cos(30^\circ) = 20.81N$$

$$F_{N_w} = \frac{mg_x}{4} = \frac{20.81N}{4} = 5.20N$$

$$F_f = \mu F_{N_w} = 1.9 \times 5.20N = 9.88N$$

Coefficient of static friction from Google

$$F_{motor} = ma = 2.45kg \times 0.15m/s^2 = 0.37N$$

$$F_{required} = F_{motor} + mg_y = 0.37N + 12.02N = 12.39N$$

$$F_{wheel} = \frac{F_{required}}{2} = \frac{12.39N}{2} = 6.20N$$

$$\tau_w = F_{wheel} \times r = 6.20N \times 0.0432m = 0.268Nm$$

Check for slippage

$$\tau_{max} = F_f \times r = 9.88N \times 0.0432m = 0.43Nm$$

$$\tau_{max} > \tau_w$$

No slippage

## 4 Checking for Tippage

$$\tau_w = 0.268Nm$$

Solve for normal forces

$$F_{net_y} = N_f + N_r - mg_y = 0$$

$$\sum M_w = N_f L - N_f L_c + F_f h = 0$$

$$N_f \times 0.135m - N_f \times 0.11m - 12.02N \times cos(30^\circ) \times 0.11m + 9.88N \times 0.01m = 8.15N$$

$$N_r = mg_y - N_r = 12.02N - 8.15N = 3.87N$$

$$h = \frac{d \times cos\theta}{sin\theta}$$

$$h_{max} = \frac{0.11m}{tan(30^\circ)} = 0.23m$$

No tippage within current design restraints

## 5 Arm Torque Requirement Calculation

$$Arm_{length} = 0.196m$$

$$Arm_{mass} = 0.127kg$$

$$Puck_{mass} = 0.17kg$$

Puck mass represents maximum weight needed to be lifted

$$\tau_{minimum} = mgl$$

$$\tau_{minimum} = (0.127kg + 0.17kg) \times 9.8m/s \times 0.196m$$

$$\tau = 0.57N$$

With a four to one gear ratio

$$GR = \frac{64}{16} = 4$$

Required torque

$$\tau_{required} = \frac{\tau_{minimum}}{GR}$$

$$\tau_{required} = \frac{0.57N}{4}$$

$$\tau_{required} = 0.143N$$

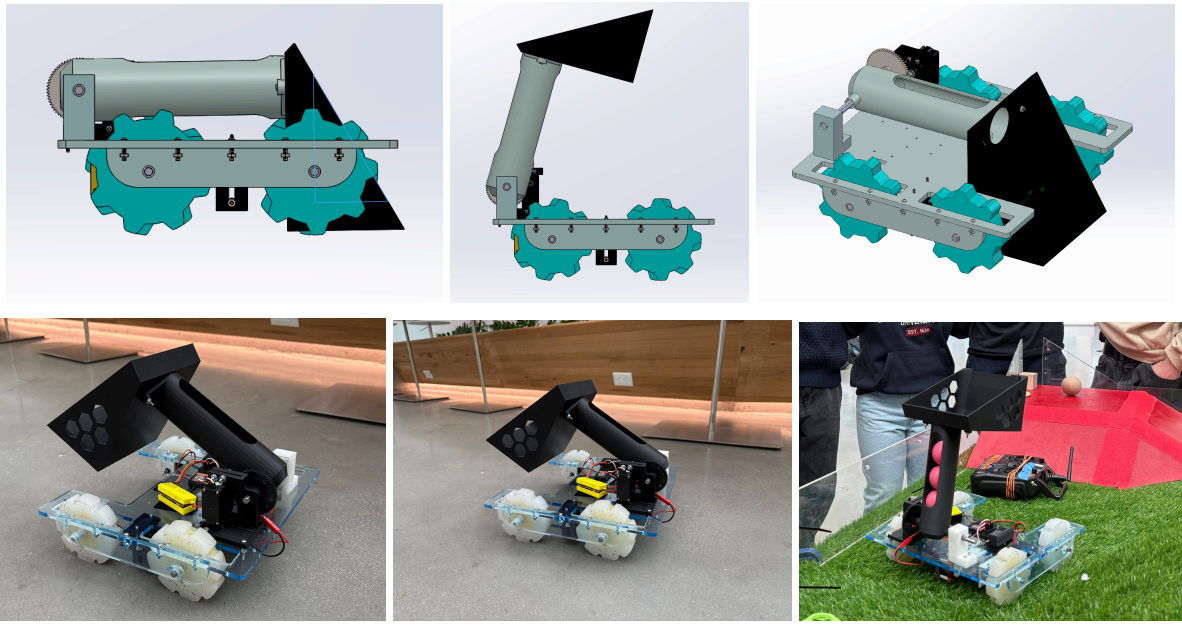
This is within servo specifications

## Final Solution

Our final solution implemented the all-in-one mechanism professor Seymur advised us to go for. Our drivetrain was made out of laser cut acrylic. For the wheels, we picked one of the designs elaborated by the members of the team and used Ecoflex 50 for a good balance of sturdiness and impact absorption. The back wheels are each powered by a DC motor dissected in class from an electric screwdriver, and the power transmission between back and front wheels happens through a mechanism of belts and 3D printed tensioners. The wheels are held in place through an acrylic piece attached to the drivetrain using t-slots. All the mechanisms related to the wheels, including the battery, are positioned under the drivetrain, as close to the ground as possible (this helps with lowering the center of gravity of the robot, preventing tippage). The mechanisms related to the arm can be found on top of the drivetrain. Those include a servo, a delrin support, a hex shaft, a gear for torque adjustment, the 3D printed arm with built-in storage and the 3D printed all purpose scoop, which included a circular hole that connected it to the arm storage through the use of the “lofted cut” tool in SolidWorks.

In the competition, our robot performed as it was designed to do. In both rounds, our group's main issue was driving the robot. We had no technical issues regarding the endurance of the robot during both rounds, except for the tensioners that became a little less tight between both rounds, but nothing that stopped the robot from working. In the first round, we had enough time to drive up and down the ramp, as well as under it, scoring a total of 25 points. In the second round, we were also able to score 25 points in the same way and due to better driving managed to find time to scoop 3 ping pong balls, scoring a total of 28 points. This punctuation wasn't enough to make us win neither rounds, but showed our robot did work for the challenges we optimized it for.

Some of the main advantages about our solution are: simplicity, universality, endurance, and reliability. The main disadvantage of our robot is the fact that it isn't able to address the stacking challenge, but we are still overall satisfied with the final product, especially because our performance in the competition was much more a result of bad driving skills than a bad design.



**Figures 8.1, 8.2, 8.3, 8.4, 8.5, and 8.6** - Final CAD and final robot in competition

Mass properties of drivetrain
Configuration: Default
Coordinate system: Coordinate System2

Mass = 5.60 pounds

Volume = 97.37 cubic inches

Surface area = 903.87 square inches

Center of mass: ( inches )
X = -0.05
Y = 1.58
Z = 2.94

Principal axes of inertia and principal moments of inertia: ( pounds * square inches )
Taken at the center of mass.
Ix = ( 1.00, -0.01, -0.01) Px = 54.57
Iy = (-0.01, -0.09, -1.00) Py = 82.95
Iz = ( 0.01, 1.00, -0.09) Pz = 123.62

Moments of inertia: ( pounds * square inches )
Taken at the center of mass and aligned with the output coordinate system. (Using posit
Lxx = 54.58 Lxy = -0.47 Lxz = -0.18
Lyx = -0.47 Lyy = 123.28 Lyz = 3.65
Lzx = -0.18 Lzy = 3.65 Lzz = 83.28

Moments of inertia: ( pounds * square inches )
Taken at the output coordinate system. (Using positive tensor notation.)
Ixx = 116.84 Ixy = -0.90 Ixz = -0.97
Iyx = -0.90 Iyy = 171.53 Iyz = 29.66
Izx = -0.97 Izy = 29.66 Izz = 97.32

**Figure 9** - SolidWorks mass properties for the robot

# Final Design Specifications and Appendix

## Specifications

| | |
|---|---|
| Robot dimension | 10.80” x 10.80” x 5.8” |
| Robot weight | 5.60 lbs |
| Turning radius | 0 in (tank design) |
| Max speed | 0.78 m/s |
| Arm lift gear ratio | 4 |
| Wheel friction | 1.8 |
| Number of motors | 2 screwdriver DC motors, 1 servo |
| Arm length | 7.71 in |
| Arm mass | 0.28 lbs |
| Scoop width | 5.3 in |
| Max height | 15 in |
| Ground clearance | 1.2 in |
| Distance between wheels | 5.05 in |
| Minimum servo torque requirement | 0.24 Nm |
| Arm base clearance | 1.53 in |
| Arm rotational freedom | 180° |

## Bill of Materials

| ITEM NO. | PART | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | Motor mount | ABS 3D printed | 2 |
| 2 | 91772A110 | Pan Head Machine Screw, 4-40 Thread, 0.5" Length | 8 |
| 3 | Motor | Motor from screwdriver | 2 |
| 4 | 92010A022 | M2.5 x 12 mm Flat Head Screw | 4 |
| 5 | 91772A116 | Pan Head Machine Screw 4-40 Thread 1.25" | 13 |
| 6 | 90126A505 Washer | Washer for Number 4 Screw Size, 0.125" ID, 0.312" OD | 18 |

| | | | |
|---|---|---|---|
| 7 | 94639A662 | Unthreaded Spacers, 3/16" OD, 11/16" Length | 8 |
| 8 | Planetary Gearbox | Planetary Gearbox from Screwdriver | 2 |
| 9 | Front Plate | Laser cut acrylic | 2 |
| 10 | Metal Gear-Shaft Interface | Metal Circle with 6 Notches from Screwdriver | 2 |
| 11 | Hex Drive Shaft | Hex Drive Shaft from Screwdriver | 2 |
| 12 | Delrin support | Machined on mill | 2 |
| 13 | Washer Screwdriver Large | Washer 0.39" ID, 0.77" OD from Screwdriver | 2 |
| 14 | 91099A169 | 18-8 Stainless Steel Phillips Flat Head Screw | 11 |
| 15 | Round Shaft | Steel axle | 2 |
| 16 | 97431A300 | E-Clip 0.25" ID | 2 |
| 17 | 95606A120 | Nylon Washer 0.25" ID | 2 |
| 18 | Washer Screwdriver Small | Washer 0.41" ID, 0.54" OD from Screwdriver | 2 |
| 19 | 98541A118 | External Retaining Ring for 10mm Shaft Diameter | 2 |
| 20 | Hex Shaft | Steel axle | 2 |
| 21 | Gear 16T 32DP Broached 025 | Acetal Plastic Gear, 16 Teeth, 32 Pitch | 2 |
| 22 | Plastic Sleeve Bearing | Nylon Flange Bushing, ID 1/4" OD 3/8" | 12 |
| 23 | Gear 48T 32DP Drilled Spring Pin | Acetal Plastic Gear, 48 Teeth, 32 Pitch | 2 |
| 24 | 92373A113 | 18-8 Stainless Steel Slotted Spring Pin | 2 |
| 25 | Wheel | Silicon molded in wax | 4 |
| 26 | Shaft Support | Aluminum | 4 |
| 27 | Pulley | 3D printed 14XL037-FS14-.5H.3L0.2500N | 2 |
| 28 | 91772A108 | Pan Head Machine Screw, 4-40 Thread, 3/8" Length | 8 |
| 29 | Base plate | Acrylic laser cut | 1 |
| 30 | Front Axle | Steel hex shaft | 2 |

| | | | |
|---|---|---|---|
| 31 | Side Joint | Acrylic laser cut | 2 |
| 32 | Tensioner | 3D printed | 2 |
| 33 | 91772A112 | 18-8 Stainless Steel Pan Head Phillips Screw | 14 |
| 34 | 90480A005 | Low-Strength Steel Hex Nut | 17 |
| 35 | Arm holder | Machined delrin | 1 |
| 36 | Servo mount | 3D printed | 1 |
| 37 | 91772A112 | 18-8 Stainless Steel Pan Head Phillips Screw | 1 |
| 38 | Arm axle | Hex steel shaft | 1 |
| 39 | 57655K13 | 14-1/2 Degree Pressure Angle Plastic Gear | 1 |

| | | | |
|---|---|---|---|
| 40 | Belt | Plastic belt | 2 |
| 41 | Arm | 3D printed | 1 |
| 42 | Scoop | 3D printed | 1 |
| 43 | Servo | SpringRC Rev1 | 1 |
| 44 | Spur gear | 32DP 64T 20PA .15FW --- S64O.5H.5L0.25N | 1 |

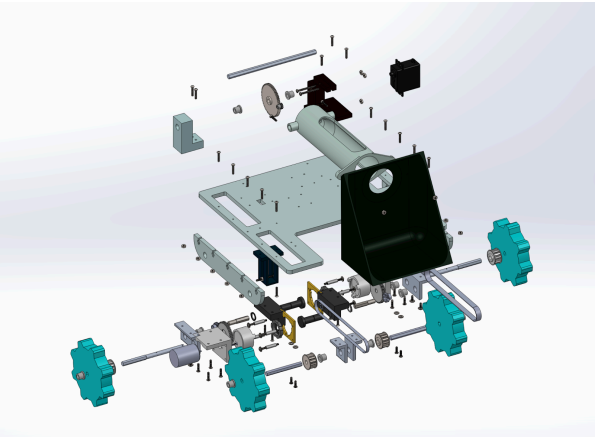


**Figure 10.1** - Exploded view

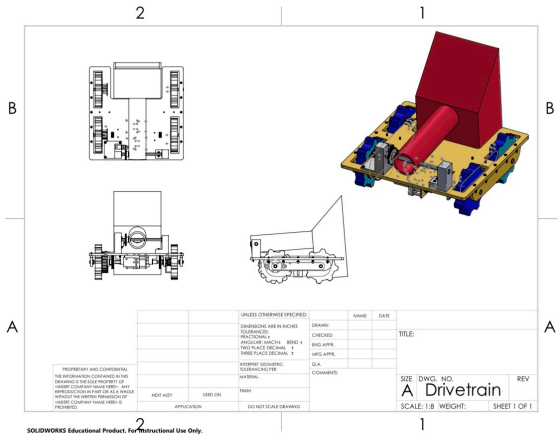



**Figure 10.2** - Assembly drawing